Baandh Prahari: a one-district pilot

बाँध प्रहरी · integration map, pilot plan, costs, data protection and roadmap

For the Water Resources Department (WRD), the Disaster Management Department (DMD) and BSDMA · October 2026 ·

Draft for discussion

This note goes with the two-page concept note. It sets out what Baandh Prahari reads from and hands back to the state's existing systems, what a pilot in one district for one monsoon would need and measure, what it would cost, how it protects people's data, and how it could grow. Anything we could not confirm from a primary source is marked **to confirm**, to be settled with the department.

1. How it works, in one incident

Report A patrol worker sends a voice note and photo on WhatsApp, or the helpline logs a call. →

Rules Published rules set P1, P2 or P3 from the sign and the river. No model decides. →

Warn For a P1, the mukhiya, ward members and subscribed families get "get ready" at once. →

Page The junior engineer on duty gets the kilometre, photo, trend and nearest team. →

Climb No answer in 10 minutes (or no delivery in 2): AE, then EE and district, then the zone's force. →

Record Every step written as it happens, locked, and handed to the district.

Relief runs on the same clock: a family's call for help goes to the nearest free relief team; a team that says "on my way" has 20 minutes before the next team is asked and the district is told.

2. Integration map: it adds to existing systems, it replaces none

Existing system	What Baandh Prahari reads	What it hands back	Permission or MoU needed
FMISC gauge map service (WRD, Patna)	Water level, danger level and trend at department and CWC gauges, every three hours[1]	Incident timelines and kilometre-level status for the department's own dashboards, if wanted	Use of the map service as an agreed feed (today it is public but not an official API)
FMISC daily flood bulletin	Gauge table (PDF), as a cross-check	None	None (public)
Embankment Asset Management System (Kosi, Bagmati reports; Kosi portal keams)[2]	Embankment lines and chainage, to replace our own kilometre marks with the department's	Incidents tagged by official chainage, building a history of weak spots	Data sharing for the Gandak (Saran) stretches. **To confirm** whether an EAMS covers the Gandak, and the name "BEAMS" used in one blueprint
RTDAS / telemetry	Live sensor levels where they exist	None	**To confirm** that such a feed exists and who runs it
CWC alerts via NDMA SACHET	CAP 1.2 river alerts, every 15 minutes[3]; an alert above danger level starts the night roll-call	None	None (public feed)
SACHET and the Cell Broadcast System (NDMA, DoT, C-DOT)	None	CAP 1.2 drafts in Hindi and English for villages behind a stretch, marked Exercise	Sending stays with the district's authorised officer, through SACHET

Flood-fighting forces (WRD resolution, 27 May 2026)[4]	The roster: 14 forces, one per zone	The zone's force is a step in the climb, between the EE and the Central Flood Control Room	The zone's roster with WhatsApp numbers
WRD helpline 1800 345 6145, district control rooms	Helpline calls, logged in a control-room form	The same clock and record for calls as for WhatsApp	Control-room staff accounts
Incident Response System (NDMA)	None	An IRS Form 001 incident briefing at hand-off to the district	The district's preferred format **to confirm**
Local Government Directory (MoPR)	LGD codes for every village and panchayat	LGD codes on every alert, timeline and relief list	None (public)
Aapda Sampoorti (DMD)	None	Villages and wards behind a breach, with LGD codes and dates, so relief verification starts sooner. This year ₹7,000 went to each of 8,27,472 families by DBT[5]	The intake format, and who receives the list, **to confirm** with DMD
Aapda Mitra volunteers, SDRF, embankment ambulances	Team positions, only from team phones on duty	Help requests nearest first, with a 20-minute clock	Team rosters. Volunteer numbers in Bihar **to confirm** (a 2025 report: 22,000 youth to be trained)[6]
JEEViKA village organisations	More reporters for danger signs	Relays for village warnings	Agreement with JEEViKA, a later phase
Sandes (NIC), government SMS	None	Pages to officials who must use a government channel; SMS to basic phones	A registered government sender (DLT); later phase

3. The pilot: Saran, monsoon 2027

Saran is where the Haijalpur breach happened this season, and where the prototype's map, gauge (Rewaghat) and villages already are. One division, one monsoon, with nothing installed in the field.

Who does what

- **WRD nodal officer:** the roster, the stretches by chainage, the escalation chain, weekly review.
- **District (DM office, control room):** receives climbs, approves any public alert, uses the briefing.
- **DMD / BSDMA:** relief teams on the map; the relief-list format.
- **Patrol workers and engineers:** save one number; a 30-minute briefing.
- **Our team:** deploys into the department's AWS account, runs drills, reports weekly, hands over the code and the account.

Timeline

Dec–Feb	MoU and data access; WhatsApp number in the department's name; AWS account
Mar–May	Load roster and chainage; train patrol workers; two dry-run drills with the district
15 Jun–31 Oct	Live season, with weekly measures to the nodal officer (the forces are posted until 31 October)[4]
November	Season report; decision on more districts

What we would measure, every week

Measure	Why it matters
Minutes from a report to the first engineer's answer (median and worst)	The gap the September review found
Share of P1 warnings answered within 10 minutes, and how many climbed	Whether the clock is respected
Roll-call coverage on high-water nights (stretches that answered "theek hai")	Whether every kilometre is watched
Households marked "ready" before the water reached them	The outcome families care about
Help requests closed safe, and minutes from request to a team on site	Whether relief reaches people in time
Unique people and villages using "Nadi ka haal"	Whether the number is already in phones on the night it matters

4. What it costs

An estimate for one district's monsoon from AWS's published prices, assuming 150 km of embankment, 150 patrol workers, 20 high-water nights with five roll-calls each, 500 reports and 150 incidents.

Item	Per season
WhatsApp messages through AWS End User Messaging Social (AWS's fee: up to $0.005 per message out, $0.001 in)	~ $115
Amazon Transcribe, Amazon Bedrock and Amazon Polly	~ $5
Step Functions, Lambda, DynamoDB, S3, Location Service, EventBridge Scheduler	~ $10
CloudWatch and Amplify Hosting	~ $25
Total	**~ $155**, plus Meta's fee on template messages

Most of the cost is roll-calls. The services are pay-per-use, so the cost from November to May is close to nothing. People's time (briefings, drills) is the larger cost and is the department's to plan. For scale, the 2024 Gandak flood damaged 118 km of embankments and cost ₹512 crore.[7]

5. Data protection and security

Personal data (DPDP Act 2023)

- The Act allows processing to assist any individual during a disaster (section 7(h)).[8] We still tell every reporter what their number is used for.
- Help requests carry no names; the family's number is stored only as a hash for de-duplication.
- Subscriptions to village warnings are kept with a consent note, for that season only.
- The public map shows counts per village; exact locations only to relief teams and the district, with a role login (a pilot adds MFA).
- Team positions only from team phones on duty. Nobody else is tracked.

Security and ownership

- Runs in AWS Mumbai (ap-south-1), a MeitY-empanelled region; data stays in India.[9]
- Logs kept 180 days, as CERT-In's 2022 directions ask.[10]
- Encryption with AWS KMS; a write-once record (S3 Object Lock); one least-privilege role per function.
- Open source (Apache-2.0), deployed by one template into the department's own AWS account. The state owns the code, the account and the data.
- No public alert is ever sent by the system: drafts are marked Exercise until the authorised officer sends them.

6. Risks, and what we would do

Risk	What we would do
It is read as blaming engineers	The same record proves a fast answer; the page hands the engineer the photo, trend and nearest team at once. No individual is named in public reports.
Phones are off or out of coverage	A page with no delivery receipt moves on in two minutes; a later phase adds government SMS and calls.
False alarms erode trust	"Get ready" asks only for cheap steps; "move" needs an engineer's confirmation or two reports from the same spot; the mukhiya can cancel a false alarm, and every cancel is logged.
A government feed is down	The last good reading is kept, and the screen says how old it is; GloFAS is the fallback for the river curve.
Voice notes in Bhojpuri or Maithili are misread	The worker's tap on the sign list sets the report; the voice note travels with it untouched.

7. Roadmap: from today's build to the whole state

Phase	What it adds	State
0. Hackathon slice (Oct 2026)	The answer clock, SACHET and FMISC in, CAP drafts out, WhatsApp, "Nadi ka haal", the public page, the live map of one stretch from AWS Open Data	**Built**
1. One-district pilot (2027)	The department's roster and chainage; WhatsApp in the department's name; relief teams on the map; IRS briefing; Hindi voice notes; geofenced check-ins	Proposed
2. The Gandak and Kosi	Embankment geometry and crest levels from the department's surveys and EAMS, replacing our drawn lines; Sentinel-1 radar water extent (sees through cloud) after every breach; government SMS for basic phones; Hyderabad standby region	**Not built**
3. Statewide	All divisions and forces; Aapda Sampoorti hand-off in its own format; JEEViKA reporters; routing on the road network around flooded areas; offline relief-camp ledgers. The reference blueprints in *docs/reference* describe these in detail.	**Not built**

A note on the reference blueprints. They were drafted with an AI tool and are treated as leads, not facts. Before any of their components is proposed, each number and system name will be checked with the department, and each AWS service checked for availability to new customers in ap-south-1. For example, AWS has closed Amazon Timestream for LiveAnalytics to new

customers,[11] so a time-series store would use another service. One blueprint places the Haijalpur breach near km 46.5 of the Saran embankment; our map uses its own kilometre marks (km 14), and the official chainage is **to confirm**. Forecasting would build on CWC's and the department's official forecasts rather than a new model.

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1. FMISC gauge map service: gis.fmiscwrdbihar.gov.in/arcgis/rest/services/FmiscHydro/FMISC_Dynmic/MapServer (tested 9 October 2026).
2. Embankment Asset Management System final reports (Kosi, Bagmati) on fmiscwrdbihar.gov.in; Kosi EAMS portal: keams.fmiscwrdbihar.gov.in.
3. SACHET Bihar feed: sachet.ndma.gov.in/cap_public_website/rss/rss_bihar.xml.
4. WRD resolution on flood-fighting forces, 27 May 2026 (state.bihar.gov.in).
5. UNI, 15 September 2026: ₹7,000 each to 8,27,472 families by DBT.
6. Patna Press, May 2025: 22,000 youth to be trained as disaster volunteers.
7. Patna Press, May 2025: the 2024 Gandak flood, 118 km of embankments damaged, ₹512 crore.
8. Digital Personal Data Protection Act 2023, section 7 (MeitY).
9. AWS compliance: MeitY empanelment of the Mumbai and Hyderabad regions.
10. CERT-In directions under section 70B, 28 April 2022.
11. AWS documentation: "Amazon Timestream for LiveAnalytics availability change".

Baandh Prahari is a student project with no government affiliation; no department has agreed to a pilot. Costs are estimates from published prices. Contact: [name, phone, email].